

Unicode Pattern Search Engine for Indian Language Wikipedia

A Unicode-Aware Text Search Engine Using Data Structures and Algorithms

Team

[Names]

[Roll Numbers]

Department & Institution

[Department]

[College Name]

ABSTRACT:

The Unicode Pattern Search Engine is designed to provide fast and efficient keyword searching in Indian language Wikipedia content. It uses algorithms such as KMP, Rabin-Karp, and Aho-Corasick along with an inverted index to search text efficiently. The system supports multiple Indian languages and provides ranked search results with relevant text snippets. This helps users quickly find useful information from large multilingual datasets.

Problem Statement

Indian language Wikipedia is expanding at an unprecedented pace, yet the tools to search it remain inadequate. Traditional substring search is too slow for large multilingual datasets, and Unicode scripts — Devanagari, Telugu, Tamil, Bengali — require specialized handling that most existing systems simply do not provide.



Core Challenge: Build a fast, scalable, Unicode-aware search engine using efficient DSA techniques.

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Rapid Content Growth

Indian language Wikipedia articles are growing rapidly, demanding scalable search infrastructure.

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Unicode Complexity

Scripts like Devanagari and Tamil require proper normalization before any search can be performed.

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Missing Capabilities

Existing systems lack efficient indexing and multi-keyword search support for Indian languages.

Key Algorithms and Data Structures

Pattern Matching Algorithms

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1

KMP

Single-pattern search with $O(n+m)$ time complexity. Avoids redundant comparisons using a failure function.

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Rabin-Karp

Hash-based multi-pattern detection. Efficient rolling hash enables simultaneous pattern scanning.

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3

Aho-Corasick

Multi-keyword matching in a single pass over the text. Ideal for large keyword sets.

Indexing Structures

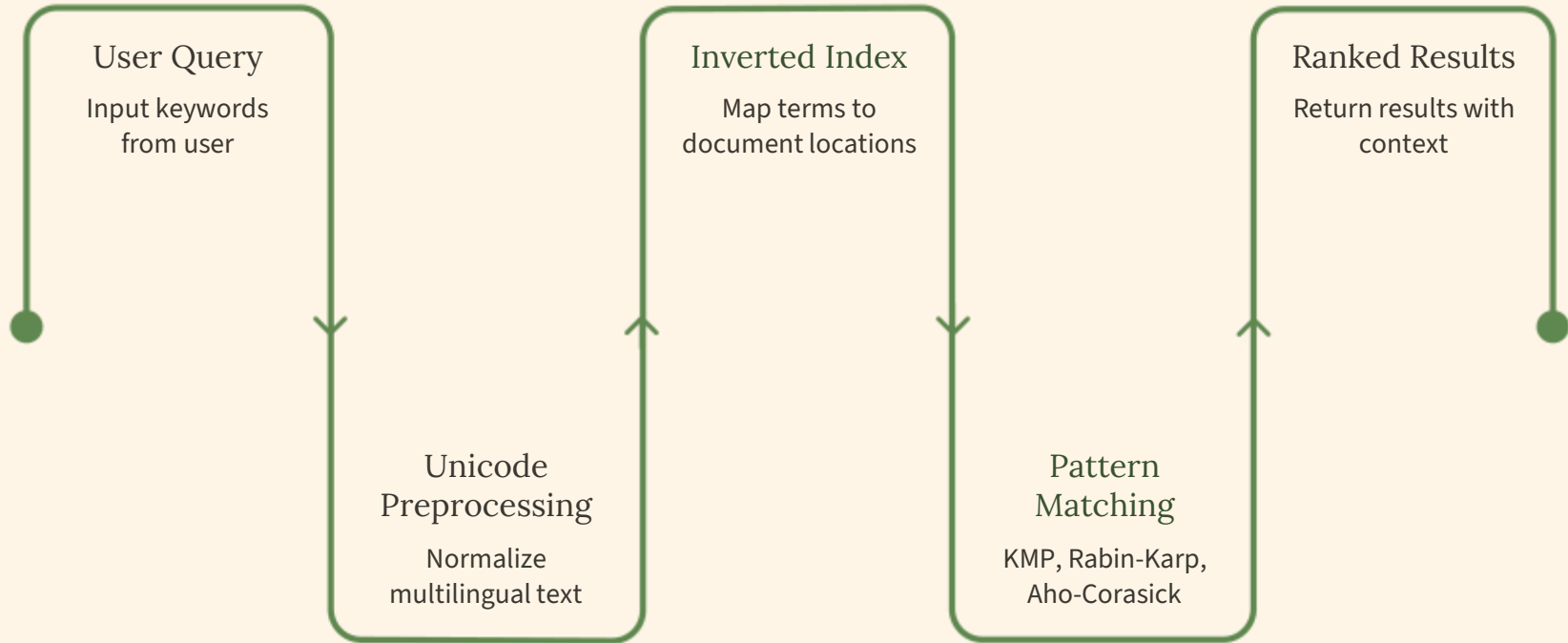
Inverted Index

Maps every keyword to the list of documents containing it, enabling sub-linear search time.

Hash Tables

Fast $O(1)$ average-case lookup and frequency counting across the document corpus.

Proposed system



Expected Outcomes & Future Scope

Expected Outcomes



Faster keyword searching in large multilingual collections



Unicode-aware support for multiple Indian languages



Efficient single and multi-keyword search



Relevant ranked results with contextual snippets

Future Scope



Advanced Ranking

Implement TF-IDF and BM25 scoring for more nuanced relevance ranking.



Web Interface

Build a web-based search interface accessible to end users across devices.



Distributed Processing

Scale to large datasets using distributed computing infrastructure.